AI learns to identify exploding stars with just 15 examples

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Diagram displaying how Google Gemini was used to analyze images and identify changes, including explosive events. Credit: Stoppa, Bulmus et al., *Nature Astronomy* (2025)

How can artificial intelligence (AI) help astronomers identify celestial objects in the night sky? This is what a recent study published in *Nature Astronomy* hopes to address as an international team of researchers investigated the potential for using AI to conduct astrophysical surveys of celestial events, including black holes consuming stars or even exploding stars themselves. This study has the potential to help astronomers use AI to enhance the field by reducing time and resources that have traditionally been used to scan the night sky.

For the study, the researchers tested Google's large language model (LLM), Gemini, on three night sky datasets: Panoramic Survey Telescope and Rapid Response System (Pan-STARRS), MeerLICHT (Dutch for "more light") and Asteroid Terrestrial-impact Last Alert System (ATLAS). The goal was to establish whether LLMs could achieve the same level of accuracy and effectiveness as the datasets listed above while presenting Gemini with three sets of images.

The researchers used specific prompts for Gemini to analyze 15 examples with instructions to classify them as "No interest," "Low interest," and "High interest" for celestial artifacts, variable stars, and explosive events, respectively, with the full repository of examples, prompts, and instructions uploaded to GitHub. The researchers then conducted a follow-up analysis six months later with Gemini having been updated with new algorithms. In the end, the researchers found that Gemini achieved an accuracy for ATLAS, MeerLICHT, and Pan-STARRS of 91.9%, 93.4%, and 94.1%, respectively.

"I've worked on this problem of rapidly processing data from sky surveys for over 10 years, and we are constantly plagued by weeding out the real events from the bogus signals in the data processing," said Dr. Stephen Smartt, who is a professor of astrophysics at the University of Oxford and a co-author on the study.

"We have spent years training machine learning models, neural networks, to do image recognition. However, the LLM's accuracy at recognizing sources with minimal guidance rather than task-specific training was remarkable. If we

can engineer to scale this up, it could be a total game changer for the field, another example of AI enabling scientific discovery."

This study comes as AI is rapidly contributing to astronomy and planetary science through a myriad of applications, including exoplanet detection, analyzing planetary surfaces and astronomical datasets, identifying supernovae, fast radio bursts, gamma-ray bursts, and gravitational waves, citizen science, theoretical modeling, and telescope operations.

An example of how AI is being used for astronomy includes the discovery of Kepler-90i, which is located approximately 2,767 light-years from Earth and is the eighth planet discovered in that system. While Kepler-90i is designated as a super-Earth at approximately 2.3 times the mass of Earth, its rocky surface temperature is far too hot to host life as we know it. Additionally, all the planets in the Kepler-90 systems orbit within the inner edge of its star's habitable zone, meaning they likely all have surfaces or atmospheres that are too hot to support life as we know it. An example of how AI is being used for planetary science includes studying Marsquakes and how seismic waves travel through the interior of Mars much differently than previously thought.

Future applications of AI in astronomy and planetary science include space weather predictions, autonomous robots on the moon and Mars, and even using AI on future crewed missions to the moon and Mars to help astronauts make better-informed decisions. Therefore, not only does this recent study demonstrate AI's increasing applications for astronomy and planetary science, but also demonstrates how non-scientists can use free online tools like Gemini to accomplish groundbreaking science.

More information: Fiorenzo Stoppa et al, Textual interpretation of transient image classifications from large language models, *Nature Astronomy* (2025). DOI: 10.1038/s41550-025-02670-z

Citation: AI learns to identify exploding stars with just 15 examples (2025, October 27) retrieved 28 October 2025 from https://phys.org/news/2025-10-ai-stars-examples.html

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